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Exploratory Data Analysis of United States Health Indicators: Trends in Life Expectancy and Behavioural Risk Factors (2000-2021)



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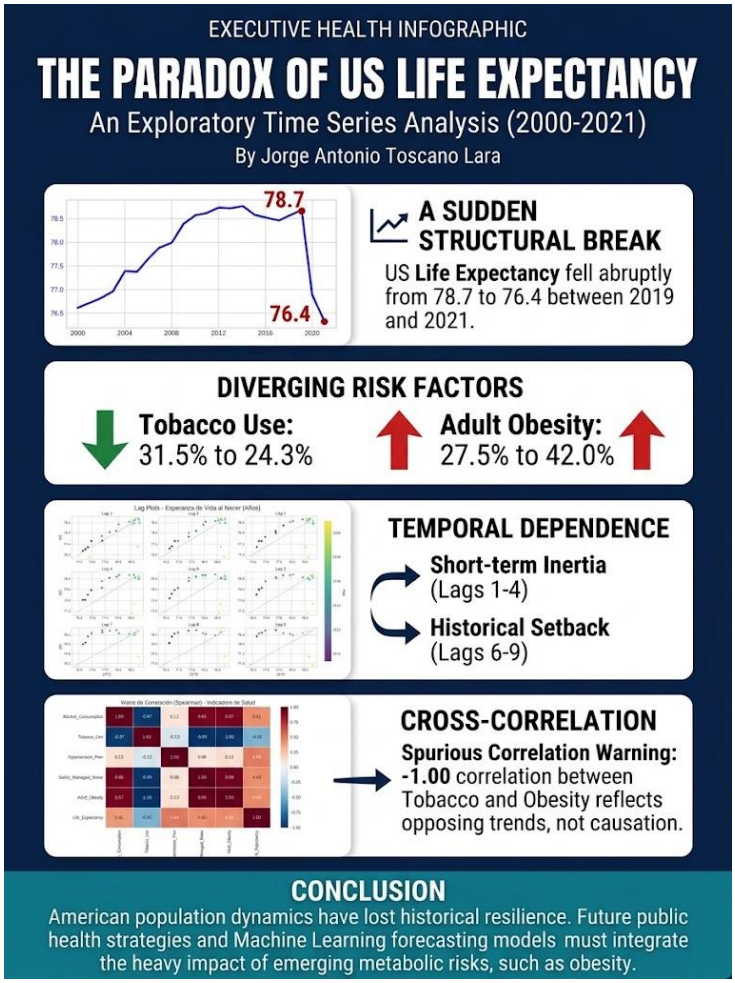
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Exploratory Data Analysis of United States Health Indicators: Trends in Life Expectancy and Behavioural Risk Factors (2000-2021)



To complement the empirical analysis and provide an immediate, executive-level synthesis of this research, a one-page infographic has been developed, this visual overview captures the vital findings of this work, highlighting the abrupt structural break in US life expectancy, the diverging trajectories between declining tobacco use and surging adult obesity, and the temporal dependencies and autocorrelation patterns identified within the dataset. It serves as a visual gateway to the rigorous exploratory analysis and policy implications discussed throughout this report.

Executive Summary

Context and Problem Definition

Life expectancy at birth is one of the most comprehensive macroeconomic and public health indicators for assessing a nation's overall well-being. Historically, developed nations have shown a consistent upward trend in this metric due to advancements in medicine, technology, and infrastructure. However, an atypical and concerning statistical phenomenon has been observed in the United States: after peaking at 78.7 years in 2019, life expectancy has experienced a period of stagnation and subsequent decline, dropping to 76.4 years.

This paradoxical behaviour suggests that underlying behavioural, environmental, or metabolic risk factors are neutralising, or even outweighing, the positive impact of medical innovations. The primary objective of this Exploratory Data Analysis (EDA) is to investigate the temporal dynamics of life expectancy in the US and its interrelation with a set of modifiable risk factors over the last two decades.

Origin and Nature of the Data

The data analysed in this report originate directly from the official records of the World Health Organization (WHO), specifically from the Global Health Observatory and the Global Health Estimates. To enable a multivariate time series analysis, multiple independent datasets were extracted and subjected to a process of cleaning, consolidation, and temporal alignment (data wrangling).

The study period spans 22 continuous years, from 2000 to 2021. Temporal interpolation techniques were applied to handle null values and irregularities in the records, thereby guaranteeing a continuous data matrix, which is an indispensable prerequisite for autocorrelation analysis and the detection of robust longitudinal trends.

Principal Analysis Variables

The analytical framework was structured around one target (dependent) variable and four contextual and risk (independent) variables:

- **Target Variable:** Life expectancy at birth for both sexes.
- **Behavioural and Metabolic Risk Factors:**
    - ***Prevalence of tobacco use:*** Percentage of the population aged 15 years and older who consume tobacco.
    - ***Alcohol consumption:*** Litres of pure alcohol consumed per capita by the population aged 15 years and older.
    - ***Adult obesity:*** Age-standardised prevalence of obesity in the population aged 18 years and older.
- **Environmental Risk Factor:**
    - ***Safely managed drinking water:*** Proportion of the population using safely managed drinking water services.

Intra-annual seasonality (months/quarters) was intentionally excluded due to the aggregated nature of the macroeconomic records, focusing the EDA on temporal dependence (lags) and long-term trends.

Relevance of the Analysis for Decision-Making

Understanding the trajectory of these data is of vital importance for the formulation of public policies, the allocation of government budgets, and the strategy of healthcare institutions.

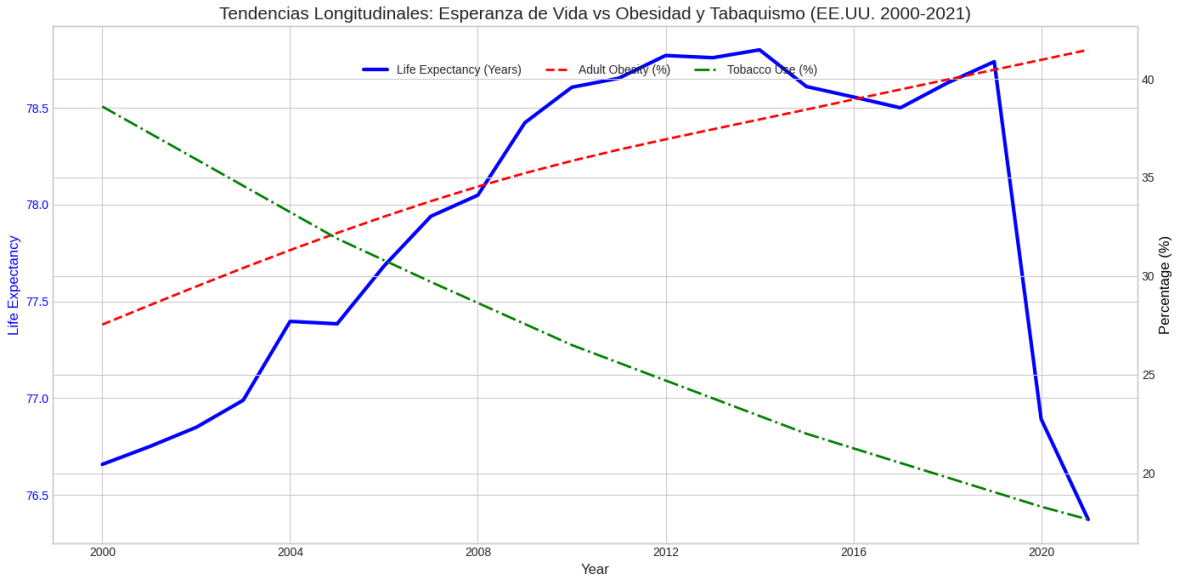
If the analysis reveals, for instance, that certain historical risks such as tobacco use have successfully decreased, but have been replaced by emerging crises like a disproportionate increase in obesity rates or alcohol consumption, decision makers must pivot their strategies. This level of data intelligence allows governments to transition from a reactive medical approach to precise preventative interventions, economically justifying the redesign of public health campaigns, food regulations, and population well-being strategies based on solid statistical evidence.

Exploratory Data Analysis (EDA)

Preliminary Methodological Considerations

Before proceeding with the visual and statistical exploration, it is fundamental to establish the temporal nature of the dataset. Comprising aggregated annual frequency observations over a 22-year period (2000-2021), a formal analysis of seasonality and cyclical behaviour is not statistically viable. Seasonality requires intra-annual data to detect repetitive fluctuations, whilst the identification of macroeconomic cycles demands considerably more extensive time horizons. Consequently, the exploration focuses on identifying long-term trends, structural breaks (outliers), temporal dependence (autocorrelation), and cross-correlation between variables.

1. Trend Analysis and Structural Breaks

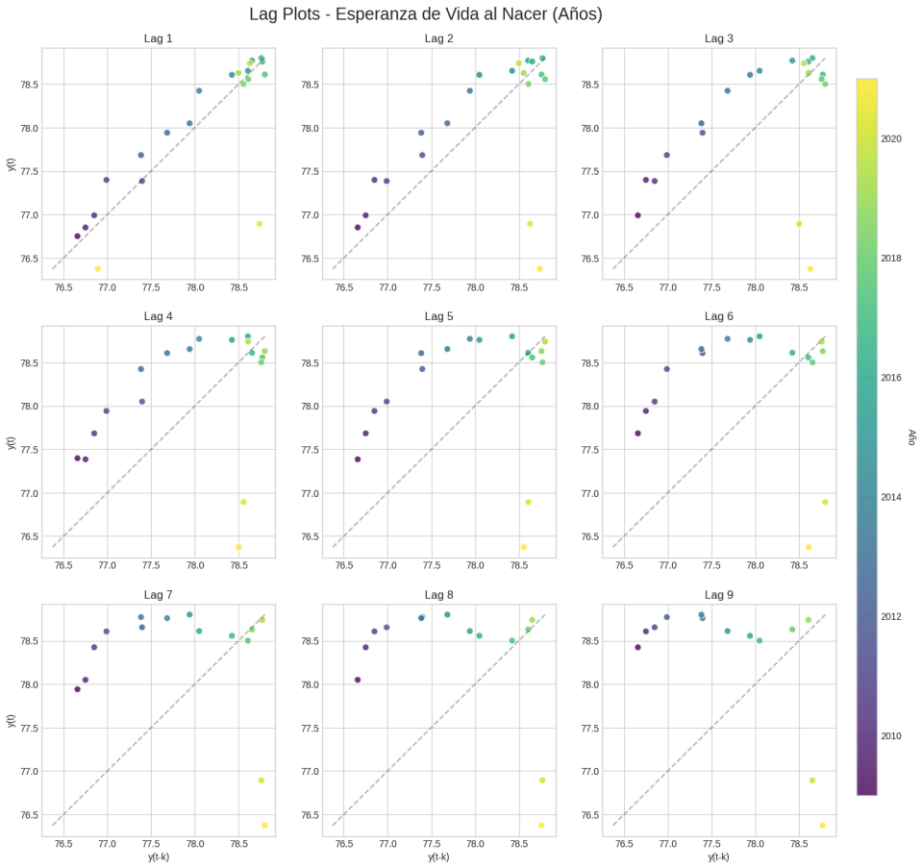


The longitudinal visualisation using multiple axes allows for contrasting the evolution of the target metric against the main risk factors. Life expectancy at birth exhibited a consistent and stable upward trend from the year 2000 (76.7 years) until reaching its historical peak in 2019 at 78.7 years. However, the time series displays a clear structural break (a generalised outlier) from 2020 onwards, culminating in an abrupt fall to 76.4 years in 2021.

Upon overlaying the behavioural risk factors, strongly directional and divergent trends are observed. On one hand, public health policies have maintained an almost linear downward trend in tobacco consumption, dropping from 31.5% in 2000 to the current 24.3%. On the other hand, the prevalence of adult obesity exhibits an alarming and continuous upward trend, climbing from 27.5% in 2000 to 42.0%. This divergence suggests that the population's metabolic deterioration could be undermining the healthcare system's resilience to external shocks.

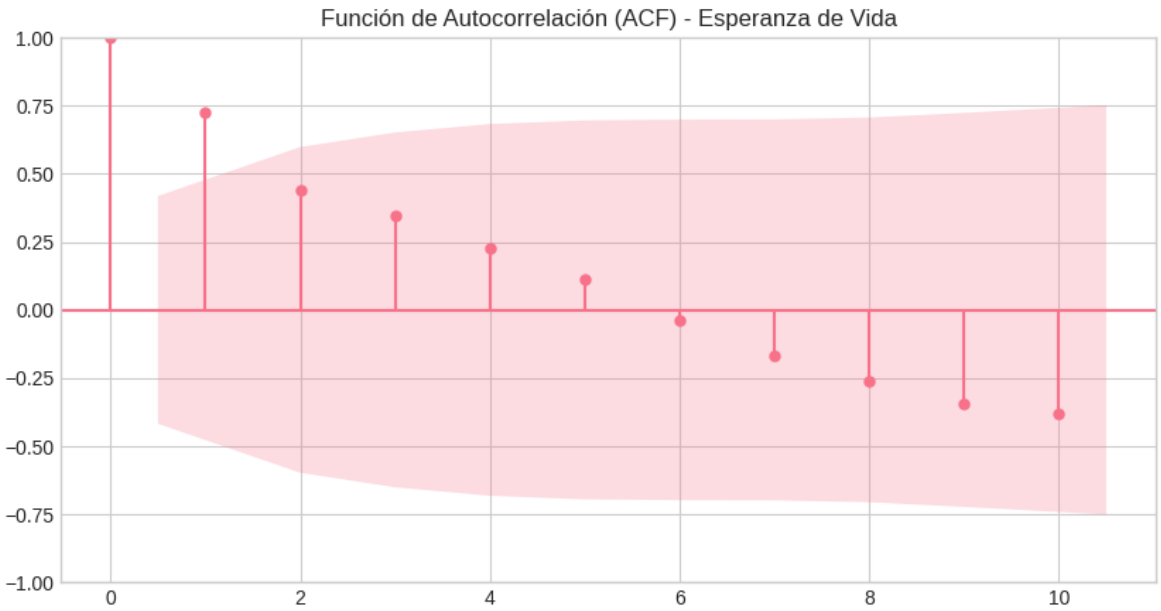
2. Temporal Dependence and Autocorrelation

To understand how past values of life expectancy influence its current state, Lag Plots and the Autocorrelation Function (ACF) were employed.



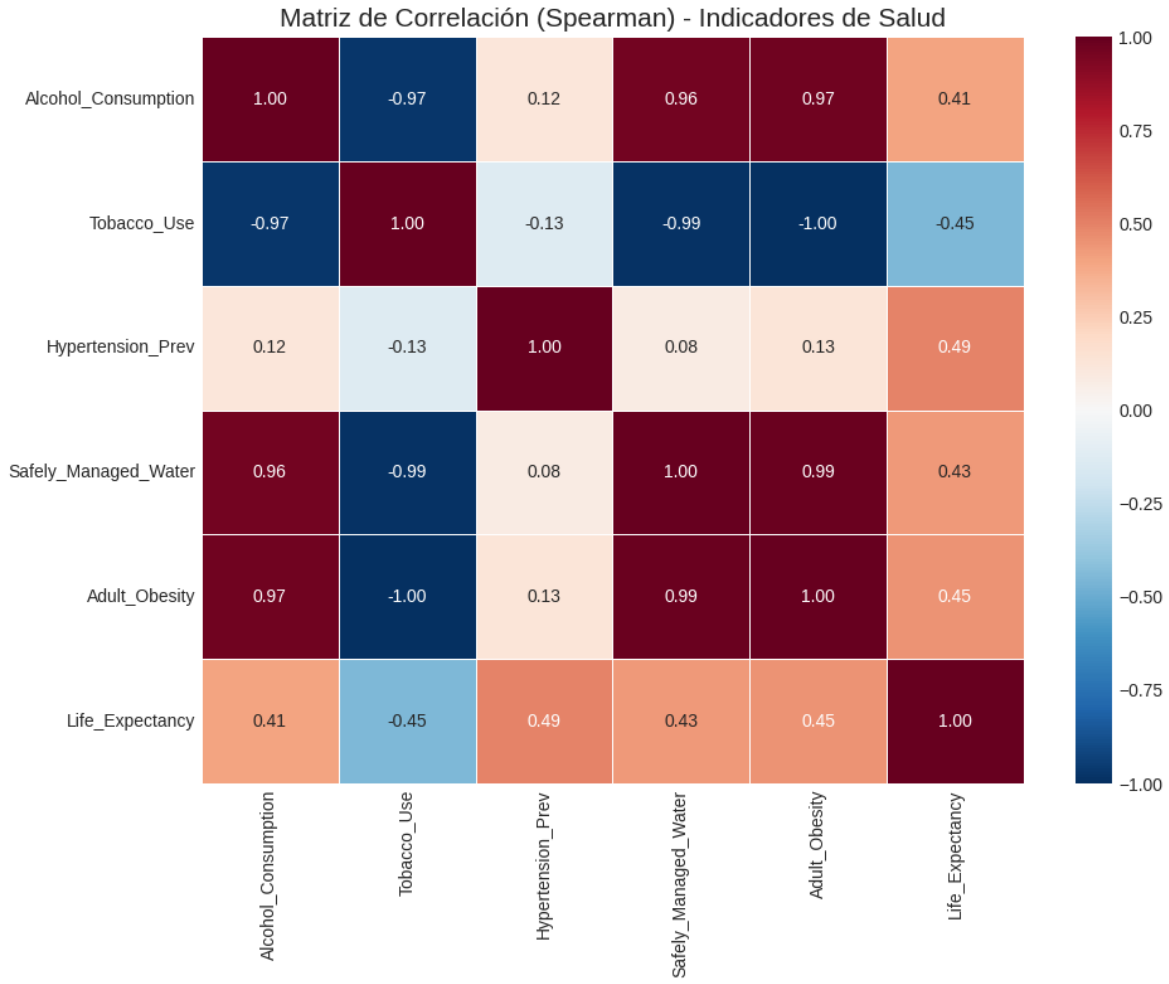
Upon analysing the topology of the Lag Plots, crucial conclusions regarding the series' dynamics are drawn:

- **Short-term inertia (Lags 1 to 4):** The points cluster very tightly around the main diagonal. This demonstrates a strong positive autocorrelation, indicating that life expectancy has a high short-term temporal dependence; the value of a given year is highly predictable and very similar to that of the immediately preceding year.
- **Historical evolution vs. Stagnation (Lags 6 to 9):** At larger lags, the majority of the points are positioned above the main diagonal. This corroborates that, historically, current life expectancy was always higher than that of previous years, reflecting the cumulative impact of technological, medical, and governmental advancements. However, the points corresponding to the most recent years (2020-2021) detach from this cluster and fall drastically. This atypical behaviour reflects a severe setback that breaks the autoregressive pattern of sustained progress that characterised the last two decades.



The ACF chart mathematically supports this observation, showing a gradual drop in correlation as the lags increase, confirming that the series is not white noise and possesses a strong trend signal.

3. Cross-Correlation Matrix (Spearman)



To evaluate the joint interaction of the variables, a correlation matrix was calculated using Spearman's rank correlation coefficient, which is robust to non-strictly linear relationships and the presence of recent outliers.

The matrix reveals complex interactions. There is a moderate correlation between life expectancy and the decrease in tobacco consumption (-0.45), suggesting a positive historical association with smoking cessation policies. However, it is imperative to approach these metrics with statistical caution. Extreme correlations are observed, such as -1.00 between tobacco use and obesity, or 0.97 between alcohol consumption and obesity. In the context of time series, this represents a clear example of spurious correlation; these variables do not necessarily have such a

perfect direct causal relationship, but rather both are governed by strong, opposing temporal trends (one consistently descending whilst the other rises uninterruptedly).

Conclusions

The Exploratory Data Analysis (EDA) applied to United States health indicators (2000-2021) reveals a panorama of critical contrasts and warns of a paradigm shift in national public health. The principal finding of this study is the statistical confirmation of a structural break in life expectancy at birth. After nearly two decades of consistent and predictable growth, driven by strong historical inertia where each year surpassed the previous one, the indicator suffered a severe setback, falling to 76.4 years by the close of the analysed period.

Visual exploration and temporal dependence analysis (Lag Plots and Autocorrelation Function) proved to be fundamental tools for understanding the magnitude of this phenomenon. These charts illustrated how life expectancy maintained a high positive autocorrelation in the short term, creating a pattern of sustained progress over the years. However, the anomalous dispersion of the most recent records (2020 and 2021) at larger lags statistically demonstrates that traditional forecasting models would fail to project the current scenario if they cannot isolate the impact of abrupt external shocks.

Furthermore, contrasting the longitudinal trends and the correlation matrix provided invaluable insight into the divergent behaviour of modifiable risk factors. Whilst public health strategies represent a resounding success in maintaining a sustained downward trend in tobacco consumption, the uninterrupted rise in adult obesity (systematically climbing to surpass the 40% threshold) emerges as a systemic crisis. The Spearman matrix yielded extreme correlations that served as a vital analytical exercise to identify spurious correlations; the EDA proves that variables with strong yet opposing trends can appear mathematically linked without necessarily obeying a direct and isolated causal relationship.

Ultimately, this dataset demonstrates that American population dynamics have lost their historical resilience. The lessons learned transcend the mere visualisation of

curves: chronic deterioration in metabolic factors is operating as a silent counterweight that diminishes the benefits of reduced tobacco use and medical advancements. For decision-making and future research, this analysis highlights the urgency of abandoning reliance on the inertia of historical growth, demanding that new public policies and Machine Learning predictive models integrate the weight of these emerging risk factors to understand the new trajectory of life expectancy.

Appendices

Appendix A: Dataset Reference

The original dataset used for this analysis was consolidated from multiple sources provided by the World Health Organization (WHO) Global Health Observatory (GHO). The raw .csv files were merged, temporally aligned by year, and filtered for the United States (2000-2021). The final pre-processed dataset (usa_health_dataset_ready_for_eda.csv) used to feed the analysis is attached to this submission.

Year	Alcohol_Consumption	Tobacco_Use	Hypertension_Prev	Safely_Managed_Water	Adult_Obesity	Life_Expectancy
2000-01-01	8.81	38.60	31.1	95.0	27.53	76.65
2001-01-01	8.81	37.25	31.3	95.0	28.51	76.75
2002-01-01	8.87	35.91	31.4	95.0	29.47	76.84
2003-01-01	8.94	34.58	31.5	95.0	30.40	76.98
2004-01-01	9.00	33.24	31.6	95.0	31.31	77.39

Appendix B: Python Script for EDA (Google Colab)

To guarantee the reproducibility and transparency of this project, the complete code used to perform the exploratory analysis (including data cleaning, autocorrelation analysis, and the generation of statistical charts) has been hosted in an interactive execution environment.

Review, and execute the complete notebook at the following Google Colab link:

https://colab.research.google.com/drive/1mJJ1qkxKpAHnBjsQuVERIAfAIOPLdtyJ?usp=sharing